Homework 9

Problem 1. *Let $f, g, h \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$.*

(a) The convolution $f * g$ is continuous and $\mathbb{Z}$ periodic. In other words $f * g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$.

(b) We have $f * g = g * f$.

(c) We have $f * (g + h) = f * g + f * h$, $\quad (f + g) * h = f * h + g * h$, and for any $c \in \mathbb{C}$, $(cf) * g = c(f * g)$.

Proof of (a). We first show that for fixed $\varepsilon > 0$, there exists a $\delta > 0$ such that for $x, x' \in \mathbb{R}$ with $|x - x'| < \delta$ then we have $\sup_{y \in [0,1]} |g(x - y) - g(x' - y)| \leq \varepsilon$. Fix some $\varepsilon > 0$. Since $g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, then it is continuous on the compact set $[0, 1]$, so by Theorem 2.3.4 we have that g is uniformly continuous on $[0, 1]$. By periodicity of g we can extend the uniform continuity to all of $\mathbb{R}$. So there exists $\delta > 0$ such that for any $u, v \in \mathbb{R}$ with $|u - v| < \delta$, then $|g(u) - g(v)| < \varepsilon$. Now fix a $y \in [0, 1]$. Then for abirtary $x, x' \in \mathbb{R}$ with $|x - x'| < \delta$, we have that $|(x - y) - (x' - y)| = |x - x'| < \delta$. So by uniform continuity we have $|g(x - y) - g(x' - y)| < \varepsilon$. Since this bound did not depend on y, then we can take the supremum over all y and still get the same bound, i.e., $\sup_{y \in [0,1]} |g(x - y) - g(x' - y)| \leq \varepsilon$.

Now we prove continuity of $f * g$. Let $\varepsilon > 0, x \in \mathbb{R}$. Note that since $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, then we have that f is bounded, that is there exists som $M > 0$ such that for all $x \in \mathbb{R}$ we have $|f(x)| < M$. By the above statement we get a $\delta > 0$ such that for all $x' \in \mathbb{R}$ with $|x - x'| < \delta$, then $\sup_{y \in [0,1]} |g(x - y) - g(x' - y)| \leq \varepsilon/(M + 1)$. Then we have for all $x' \in \mathbb{R}$ such that $|x - x'| < \delta$, we have

$$\begin{aligned}
|(f * g)(x) - (f * g)(x')| &= \left| \int_0^1 f(y) g(x - y) dy - \int_0^1 f(y) g(x' - y) dy \right| \\
&= \left| \int_0^1 f(y)(g(x - y) - g(x' - y)) dy \right| \\
&\leq \int_0^1 |f(y)||g(x - y) - g(x' - y)| dy \\
&\leq \sup_{y \in [0,1]} |g(x - y) - g(x' - y)| \int_0^1 |f(y)| dy \\
&\leq \frac{\varepsilon}{M + 1} \cdot M < \varepsilon.
\end{aligned}$$

This shows continuity. Now we show that $f * g$ is $\mathbb{Z}$ periodic. Let $k \in \mathbb{Z}$. Then we have

$$\begin{aligned}
(f * g)(x + k) &= \int_0^1 f(y) g(x + k - y) dy \\
&= \int_0^1 f(y) g(x - y) dy \qquad \text{(periodicity of } g\text{)}
\end{aligned}$$

$$= (f * g)(x).$$

We are done. □

Proof of (b). Note that $(f * g)(x) = \int_0^1 f(y)g(x-y)dy$. Let $y' = x - y$. Then $dy = -dy'$. So we have $\int_0^1 f(y)g(x-y)dy = \int_x^{x-1} -g(y')f(x-y')dy'$. By periodicity we have $\int_{x-1}^x g(y')f(x-y')dy = \int_0^1 g(y')g(x-y')dy = (g * f)(x)$. □

Proof of (c). We have

$$\begin{aligned}
(f * (g+h))(x) &= \int_0^1 f(y)[(g+h)(x-y)]dy \\
&= \int_0^1 f(y)[g(x-y) + h(x-y)]dy \\
&= \int_0^1 f(y)g(x-y)dy + \int_0^1 f(y)h(x-y)dy \\
&= (f * g)(x) + (f * h)(x).
\end{aligned}$$

From the above derivation and (b) we have

$$((f+g) * h)(x) = (h * (f+g))(x) = (h * f)(x) + (h * g)(x) = (f * h)(x) + (g * h)(x).$$

Now let $c \in \mathbb{C}$. We have,

$$\begin{aligned}
c(f * g)(x) &= c\int_0^1 f(y)g(x-y)dy \\
&= \int_0^1 (cf)(y)g(x-y)dy \\
&= ((cf) * g)(x).
\end{aligned}$$

Symmetrically we have

$$\begin{aligned}
c(f * g)(x) &= c\int_0^1 f(y)g(x-y)dy \\
&= \int_0^1 f(y)[(cg)(x-y)]dy \\
&= (f * (cg))(x).
\end{aligned}$$

□

Problem 2. *Let f be a function in $C(\mathbb{R}/\mathbb{Z};\mathbb{C})$, and define the trigonometric Fourier coefficients a_n, b_n for $n = 0, 1, 2, ...$ by*

$$a_n := 2\int_0^1 f(x)\cos(2\pi nx)dx; \quad b_n := \int_0^1 f(x)\sin(2\pi nx)dx.$$

(a) Show that the series $1/2a_0 + \sum_{n=1}^{\infty} a_n\cos(2\pi nx) + b_n\sin(2\pi nx)$ converges to f in the L^2 metric.

(b) Show that if $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent, then the above series actually converges uniformly to f, and not just in L^2 metric

Proof of (a). By the Forier Theorem we have that the series $\sum_{n=-\infty}^{\infty} \hat{f}(n)e_n$ converges to f in the L^2 metric. For simplicity sake denote $I^n_{\cos} := \int_0^1 f(x)\cos(2\pi nx)dx$ and $I^n_{\sin} := \int_0^1 f(x)\sin(2\pi nx)dx$. By definition we have $\hat{f}(n)e_n = \langle f, e_n\rangle e_n = e^{2\pi inx}\int_0^1 f(x)e^{-2\pi nx}dx$. We shall expand this integral into sines and cosines. We have,

$$\begin{aligned}\hat{f}(n)e_n &= (\cos(2\pi nx) + i\sin(2\pi nx))\int_0^1 f(x)(\cos(2\pi nx) - i\sin(2\pi nx))dx \\ &= (\cos(2\pi nx) + i\sin(2\pi nx)(I^n_{\cos} - iI^n_{\sin}) \\ &= \cos(2\pi nx)I^n_{\cos} - i\cos(2\pi nx)I^n_{\sin} + i\sin(2\pi nx)I^n_{\cos} + \sin(2\pi nx)I^n_{\sin}.\end{aligned}$$

Now we can split the infinite sum into three parts:

$$\sum_{n=-\infty}^{\infty} \hat{f}e_n = \sum_{n=-\infty}^{-1} \hat{f}(n)e_n + \hat{f}(0)e_0 + \sum_{n=1}^{\infty} \hat{f}(n)e_n.$$

In the sum from $-\infty$ to -1 all $n < 0$, so in particular we have

$$\sum_{n=-\infty}^{-1} \hat{f}(n)e_n = \sum_{n=1}^{\infty} \hat{f}(-n)e_{-n}.$$

In the expansion of $\hat{f}(-n)e_{-n}$, the first term stays the same since cos is even, the second term gets a negative sign since cos even and sin odd, the third term also gets a negative sign by the same logic, and the last term stays the same since both sin is odd and they are being multiplied. So we get

$$\begin{aligned}\sum_{n=-\infty}^{-1} \hat{f}(n)e_n &= \sum_{n=1}^{\infty} \hat{f}(-n)e_{-n} \\ &= \sum_{n=1}^{\infty}[\cos(2\pi nx)I^n_{\cos} + i\cos(2\pi nx)I^n_{\sin} - i\sin(2\pi nx)I^n_{\cos} + \sin(2\pi nx)I^n_{\sin}].\end{aligned} \tag{1}$$

For the positive sum we can keep it the same. We have,

$$\sum_{n=1}^{\infty} \hat{f}(n)e_n = \sum_{n=1}^{\infty}[\cos(2\pi nx)I^n_{\cos} - i\cos(2\pi nx)I^n_{\sin} + i\sin(2\pi nx)I^n_{\cos} + \sin(2\pi nx)I^n_{\sin}]. \tag{2}$$

Adding equation (1) and (2) yields

$$\begin{aligned}\sum_{n=-\infty}^{\infty} \hat{f}(n)e_n &= \hat{f}(0)e_0 + \sum_{n=-\infty}^{-1} \hat{f}(n)e_n + \sum_{n=1}^{\infty} \hat{f}(n)e_n \\ &= \hat{f}(0)e_0 + \sum_{n=1}^{\infty} 2\cos(2\pi nx)I_{\cos}^n + 2\sin(2\pi nx)I_{\sin}^n.\end{aligned}$$

Note that

$$\hat{f}(0)e_0 = \langle f, e_0\rangle e_0 = e^0 \int_0^1 f(x)e^0 dx = \int_0^1 f(x) = \int_0^1 f(x)\cos(2\pi(0)x)dx = \frac{1}{2}a_0$$

. Since $a_n = 2I_{\cos}^n$ and $b_n = 2I_{\sin}^n$ we get,

$$\sum_{n=-\infty}^{\infty} \hat{f}(n)e_n = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \cos(2\pi nx)a_0 + \sin(2\pi nx)b_0.$$

We are done. $\square$

Proof of (b). By Theorem 5.5.3 if we show that $\sum_{n=-\infty}^{\infty}\left|\hat{f}(n)\right|$, then we have that $\sum_{n=-\infty}^{\infty}\hat{f}(n)e_n$ uniformly converges to f. So by equality we will have that the series in (a) will also coverge uniformly to f. As done in (a) we have that $\sum_{n=-\infty}^{\infty}|\hat{f}(n)| = |\hat{f}(0)| + \sum_{n=-\infty}^{-1}|\hat{f}(n)| + \sum_{n=1}^{\infty}|\hat{f}(n)|$. Since $\hat{f}(0)$ is just a constant, then if we show that $\sum_{n=-\infty}^{-1}|\hat{f}(n)|$ and $\sum_{n=1}^{\infty}|\hat{f}(n)|$ both converge absolutely then we are done. We can write

$$\begin{aligned}\sum_{n=-\infty}^{-1}|\hat{f}(n)| &= \sum_{n=1}^{\infty}|\hat{f}(-n)| \\ &= \sum_{n=1}^{\infty}\left|\int_0^1 f(x)(\cos(2\pi nx) - i\sin(2\pi nx))dx\right| \\ &= \sum_{n=1}^{\infty}\left|\int_0^1 f(x)\cos(2\pi nx)dx - i\int_0^1 f(x)\sin(2\pi nx)dx\right| \\ &\overset{\triangle}{\leq} \sum_{n=1}^{\infty}\left|\int_0^1 f(x)\cos(2\pi nx)dx\right| + \left|\int_0^1 f(x)\sin(2\pi nx)\right| \\ &= \sum_{n=1}^{\infty}\frac{1}{2}|a_n| + \frac{1}{2}|b_n|.\end{aligned}$$

Note that by assumption $\sum_{n=1}^{\infty}|a_n|$ and $\sum_{n=1}^{\infty}|b_n|$ coverge absolutely, so by comparison we have that $\sum_{n=-\infty}^{-1}\left|\hat{f}(n)\right|$ also converges absolutely. The proof for why $\sum_{n=1}^{\infty}\left|\hat{f}(n)\right|$ converges is symmetric to the one done above. So we are done. $\square$

Problem 3. *Let $f(x)$ be the function defined by $f(x) = (1-2x)^2$ when $x \in [0,1)$, and extend to be $\mathbb{Z}$ periodic for the rest of the real line.*

(a) Using exercise 5.5.1, show that $1/3 + \sum_{n=1}^{\infty}\frac{4}{\pi^2 n^2}\cos(2\pi nx)$ converges uniformly to f.

(b) Conclude that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.
(c) Conclude that $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^2}{90}$.

Proof of (a). We shall find a_n and b_n from the previous problem. For a_n we have

$$2\int_0^1 (1-2x)^2 \cos(2\pi nx)dx = \underbrace{2\int_0^1 \cos(2\pi nx)}_{(1)} - \underbrace{8\int_0^1 x\cos(2\pi nx)dx}_{(2)} + \underbrace{8\int_0^1 x^2\cos(2\pi nx)dx}_{(3)}.$$

We will solve (1), (2), and (3) independently. For (1) we have,

$$2\int_0^1 \cos(2\pi nx)dx = 2\left[\frac{\sin(2\pi nx)}{2\pi n}\right]_0^1 = 0.$$

For (2) we have,

$$\begin{aligned}
8\int_0^1 x\cos(2\pi nx)dx &= 8\int_0^1 xd\left(\frac{\sin(2\pi nx)}{2\pi n}\right) \\
&\overset{\text{IBP}}{=} 8\left(\frac{x\sin(2\pi nx)}{2\pi n}\,\Bigg|_0^1 - \int_0^1 \frac{\sin(2\pi nx)}{2\pi n}dx\right) \\
&= 8(0-0) \\
&= 0.
\end{aligned}$$

For (3) we have,

$$\begin{aligned}
8\int_0^1 x^2\cos(2\pi nx) &= 8\int_0^1 x^2 d\left(\frac{\sin(2\pi nx)}{2\pi n}\right) \\
&= 8\left(\frac{x^2\sin(2\pi nx)}{2\pi n}\,\Bigg|_0^1 - \int_0^1 2x\frac{\sin(2\pi nx)}{2\pi n}dx\right) \\
&= 8\left(0 - \frac{1}{\pi n}\int_0^1 xd\left(-\frac{\cos(2\pi nx)}{2\pi n}\right)\right) \\
&= -\frac{8}{\pi n}\left(\frac{-x\cos(2\pi nx)}{2\pi n}\,\Bigg|_0^1 + \frac{1}{2\pi n}\int_0^1 \cos(2\pi nx)dx\right) \\
&= -\frac{8}{\pi n}\left(-\frac{1}{2\pi n} + 0\right) \\
&= \frac{4}{\pi^2 n^2}.
\end{aligned}$$

So we have that $a_n = 4/(\pi^2 n^2)$. We will now find b_n. We have,

$$2\int_0^1 (1-2x)^2\sin(2\pi nx)dx = \underbrace{2\int_0^1 \sin(2\pi nx)}_{(1)} - \underbrace{8\int_0^1 x\sin(2\pi nx)dx}_{(2)} + \underbrace{8\int_0^1 x^2\sin(2\pi nx)dx}_{(3)}.$$

Similaryly we will solve (1), (2), and (3) independently. For (1) we have,

$$2\int_0^1 \sin(2\pi nx) = -2\left[\frac{\cos(2\pi nx)}{2\pi n}\right]_0^1 = 0.$$

For (2) we have,

$$\begin{aligned}
-8\int_0^1 x\sin(2\pi nx)dx &= 4\int_0^1 xd\Big(-\frac{\cos(2\pi nx)}{2\pi n}\Big) \\
&= -8\left(\frac{-x\cos(2\pi nx)}{2\pi n}\ \Big|_0^1 + \int_0^1 \frac{\cos(2\pi nx)}{2\pi n}dx\right) \\
&= -8\Big(-\frac{1}{2\pi n}+0\Big) \\
&= \frac{4}{2\pi n}.
\end{aligned}$$

For (3) we have,

$$\begin{aligned}
8\int_0^1 x^2\sin(2\pi nx)dx &= 8\int_0^1 x^2 d\Big(-\frac{\cos(2\pi nx)}{2\pi n}\Big) \\
&= 8\left(\frac{-x^2\cos(2\pi nx)}{2\pi n}\ \Big|_0^1 + \underbrace{\int_0^1 2x\frac{\cos(2\pi nx)}{2\pi n}dx}_{0 \text{ from } a_n \text{ calculation}}\right) \\
&= 8\Big(-\frac{1}{2\pi n}+0\Big) \\
&= -\frac{4}{2\pi n}.
\end{aligned}$$

So we have that $b_n = \frac{4}{2\pi n} - \frac{4}{2\pi n} = 0$. Now we need a_0. We have

$$a_0 = 2\int_0^1 (1-2x)^2dx = \left[-\frac{(1-2x)^3}{3}\right]_0^1 = \frac{2}{3}.$$

Since both

$$\sum_{n=1}^{\infty}|a_n| = \sum_{n=1}^{\infty}\left|\frac{4}{\pi^2n^2}\right| \text{ and } \sum_{n=1}^{\infty}|b_n| = \sum_{n=1}^{\infty}0$$

both converge absolutely, then by problem 2, we have that

$$\frac{1}{2}\cdot\frac{2}{3}+\sum_{n=1}^{\infty}\frac{4}{\pi^2n^2}\cos(2\pi nx)=\frac{1}{3}+\sum_{n=1}^{\infty}\frac{4}{\pi^2n^2}\cos(2\pi nx)$$

converges uniformly to $(1-2x)^2$. □

Proof of (b). We have $f(x)=(1-2x)^2=\frac{1}{3}+\sum_{n=1}^{\infty}\frac{4}{\pi^2n^2}\cos(2\pi nx)$ where $x\in[0,1)$. For $x=0$ we have

$$1=\sum_{n=1}^{\infty}\frac{4}{\pi^2n^2}\implies\frac{\pi^2}{4}=\sum_{n=1}^{\infty}\frac{1}{n^2}.$$

□

Proof of (c). We can expand the series into its exponential parts. We have

$$\begin{aligned}\frac{1}{3}+\sum_{n=1}^{\infty}\frac{4}{\pi^2n^2}\cos(2\pi nx)&=\frac{1}{3}+\sum_{n=1}^{\infty}\frac{4}{2\pi^2n^2}\frac{e_n+e_{-n}}{2}\\&=\frac{1}{3}+\sum_{n=1}^{\infty}\frac{2}{\pi^2n^2}e_n+\sum_{n=1}^{\infty}\frac{2}{\pi^2n^2}e_{-n}\\&=\frac{1}{3}+\sum_{n=1}^{\infty}\frac{2}{\pi^2n^2}e_n+\sum_{n=-\infty}^{-1}\frac{2}{\pi^2n^2}e_n.\end{aligned}$$

So then $\hat{f}(n)=\frac{2}{\pi^2n^2}$ when $n\neq 0$ and $\hat{f}(0)=\frac{1}{3}$. Now we find $\|f\|_2^2$. We have,

$$\begin{aligned}\|f\|_2^2&=\langle f,f\rangle\\&=\big\langle(1-2x)^2,(1-2x)^2\big\rangle\\&=\int_0^1(1-2x)^4dx\\&=-\frac{1}{10}(1-2x)^5\Big|_0^1\\&=\frac{1}{5}.\end{aligned}$$

So by Plancherel's Theorem we have,

$$\begin{aligned}\frac{1}{5}&=\frac{1}{9}+\sum_{n=1}^{\infty}\frac{4}{\pi^4n^4}+\sum_{n=-\infty}^{-1}\frac{4}{\pi^4n^4}\\\frac{4}{45}&=\sum_{n=1}^{\infty}\frac{4}{\pi^4n^4}+\sum_{n=1}^{\infty}\frac{4}{\pi^4n^4}\\\frac{4}{45}&=2\sum_{n=1}^{\infty}\frac{4}{\pi^4n^4}\\\frac{\pi^4}{90}&=\sum_{n=1}^{\infty}\frac{1}{n^4}.\end{aligned}$$

$\square$

Problem 4. *If $f \in C(\mathbb{R}/\mathbb{Z};\mathbb{C})$ and P is a trigonometric polynomial, show that*

$$\widehat{f*p}(n) = \hat{f}(n)c_n = \hat{f}(n)\hat{P}(n)$$

for all integers n. More generally, if $f, g \in C(\mathbb{R}/\mathbb{Z};\mathbb{C})$, show that

$$\widehat{f*g}(n) = \hat{f}(n)\hat{g}(n)$$

for all integers n.

Proof. Let $P(x) = \sum_{m=-M}^{M} c_m e_m$. Then we want to find $\widehat{f*P}(n)$. Note by definition we have $\widehat{f*P}(n) = \langle f*P, e_n\rangle$. Let this be equation (1). We will simplify $f*P(x)$ first. We have,

$$\begin{aligned} f*P(x) &= \left(f * \sum_{m=-M}^{M} c_m e^{2\pi i m x}\right)(x) \\ &= \int_0^1 f(y) \sum_{m=-M}^{M} c_m e^{2\pi i m(x-y)} dy \\ &= \sum_{m=-M}^{M} c_m e^{2\pi i m x} \int_0^1 f(y) e^{-2\pi i m y} dy \\ &= \sum_{m=-M}^{M} c_m e^{2\pi i m x} \hat{f}(m). \end{aligned}$$

Plugging this result into (1) gives,

$$\begin{aligned} \widehat{f*P}(n) &= \int_0^1 \sum_{m=-M}^{M} c_m e^{2\pi m x} \hat{f}(m) e^{-2\pi i n x} dx \\ &= \sum_{m=-M}^{M} c_m \hat{f}(m) \int_0^1 e^{2\pi i (m-n)x} dx \\ &= \sum_{m=-M}^{M} c_m \hat{f}(m) \langle e_m, e_n\rangle \end{aligned}$$

where $\langle e_m, e_n\rangle = 1$ if $m = n$ and 0 otherwise. So if $|n| \le M$ then we have that

$$\widehat{f*P}(n) = \sum_{m=-M}^{M} c_m \hat{f}(m)\langle e_m, e_n\rangle = \begin{cases} c_n \hat{f}(m) & \text{if } |n| \le M \\ 0 & \text{else} \end{cases}$$

Note that

$$\hat{P}(n) = \langle P, e_n\rangle = \int_0^1 \sum_{m=-M}^{M} c_m e_m e_{-n} dx = \sum_{m=-M}^{M} c_m \int_0^1 e_m e_{-n} dx = \sum_{m=-M}^{M} c_m \langle e_m, e_n\rangle = \begin{cases} c_n \text{ if } |n| \le M \\ 0 \text{ else} \end{cases}.$$

So then we can write

$$\widehat{f * P}(n) = \hat{f}(n)\hat{P}(n)$$

for all $n \in \mathbb{Z}$.

Now we prove the general case. Let $g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. By Weierstrass approximation theorem we have that there exists a sequence of trigonometric polynomials $(P_N)_{N\in\mathbb{N}}$ such that $\|g - P_N\|_\infty \longrightarrow 0$. So for fixed $N \in \mathbb{N}$ we have that $\widehat{f * P_N}(n) = \hat{f}(n)\widehat{P_N}(n)$ by the first part. If we can show that as $N \to \infty$ we have that $\widehat{f * P_N}(n) \longrightarrow \widehat{f * g}(n)$ and $\hat{f}(n)\widehat{P_N}(n) \longrightarrow \hat{f}(n)\hat{g}(n)$ then we are done. We will show each side separately.

Lemma: If $h_k \longrightarrow h$ uniformly, then $\widehat{h_k}(n) \longrightarrow \hat{h}(n)$ for fixed $n \in \mathbb{Z}$.

Let $\varepsilon > 0$. Since $h_k \longrightarrow h$ uniformly, then there is a K such that for all $k > K$ and $x \in \mathbb{R}$ we have that $\|h_k(x) - h(x)\|_\infty < \varepsilon$. Notice,

$$\begin{aligned}
\left|\widehat{h_k}(n) - \hat{h}(n)\right| &= \left|\int_0^1 h_k(x)e^{-2\pi inx}dx - \int_0^1 h(x)e^{-2\pi inx}dx\right| \\
&= \left|\int_0^1 (h_k(x) - h(x))e^{-2\pi inx}dx\right| \\
&\leq \int_0^1 |h_k(x) - h(x)|\underbrace{\left|e^{-2\pi inx}\right|}_{=1}dx \\
&= \int_0^1 |h_k(x) - h(x)|dx \\
&\leq \|h_k(x) - h(x)\|_\infty < \varepsilon.
\end{aligned}$$

Claim: $f * P_N \longrightarrow f * g$ uniformly. Indeed we have,

$$\begin{aligned}
|f * P_N(x) - f * g| &= \left|\int_0^1 f(y)P_N(x-y)dy - \int_0^1 f(x)g(x-y)dy\right| \\
&= \left|\int_0^1 f(y)(P_N(x-y) - g(x-y))dy\right| \\
&\leq \int_0^1 |f(y)||P_N(x-y) - g(x-y)|dy \\
&\leq \|f\|_\infty \|P_N - g\|_\infty.
\end{aligned}$$

Since f bounded and $P_N \longrightarrow g$ uniformly then we have proven the claim. So we can apply $f * P_N$ to our lemma above. We have that $\widehat{f * P_N} \longrightarrow \widehat{f * g}$. Now since $P_N \longrightarrow g$ uniformly, then from the lemma we have $\widehat{P_N} \longrightarrow \hat{g}$. Hence we have $\hat{f}(n)\widehat{P_N}(n) \longrightarrow \hat{f}(n)\hat{g}(n)$. So we are done.

$\square$